Executive Audit Verdict & Strategic Direction

> **Recommendation:** The directional forecasting component of the PRE Pre-Earnings Forecast Model should be formally retired.
> The model exhibits zero directional edge, generating a 51.9% hit rate that lags the simple baseline of predicting "ALWAYS DOWN" (55.6%), with a mean idiosyncratic separation of -1.86pp (t = -0.50). Due to earnings date clustering, the effective sample size is n = 6 independent event-dates, not n = 27. Under realistic market volatility, detecting a viable directional edge would require approximately 784 graded events (~3 years of live exposure).

However, the underlying pre-registration discipline, clock guards, primary filing verification, and forensic audit procedures are demonstrably high-value. PRE should be rescoped from a predictive directional model into an automated **Earnings Quality & Forensic Audit Engine**, alongside an optional continuous probability model for volatility/straddle trading.

Section 1: Statistical Audit & Independent Verification

1.1 Core Metric Re-Derivation

Independent re-computation from raw prompt data confirms all primary statistical findings:

- **Model Hit Rate:** 14 / 27 = 51.85% (approx. 51.9%)
- **ALWAYS DOWN Benchmark:** 15 / 27 = 55.56% (approx. 55.6%) — Edge vs Benchmark: -3.71pp
- **ALWAYS UP Benchmark:** 12 / 27 = 44.44% (approx. 44.4%)
- **Mean Idiosyncratic Return (Model = UP, n=11):** -0.85% (std dev = 8.24%)
- **Mean Idiosyncratic Return (Model = DOWN, n=16):** +1.02% (std dev = 11.06%)
- **Mean Separation (Δμ):** -0.85% - (+1.02%) = -1.87pp (approx. -1.86pp)

Standard Error & t-Statistic Calculation

The standard error of separation between the two independent sample means is:

```
SE = sqrt( (8.24^2 / 11) + (11.06^2 / 16) )
   = sqrt( 6.1725 + 7.6452 )
   = sqrt( 13.8177 ) ≈ 3.717%
```

Evaluating the t-statistic:

```
t = (-1.87) / 3.717 ≈ -0.503  (p ≈ 0.62)
```

With t = -0.503, the directional separation is statistically indistinguishable from zero noise.

1.2 Clustering Fallacy & Sample Size Analysis

Evaluating n = 27 stock prints across 6 calendar dates treats dependent observations as independent:

Event Date	Ticker Count (n)	Correct Calls	Accuracy Rate	Sector / Macro Overlay
2026-07-30	12	7	58.3%	Tech / Mega-cap cluster
2026-07-31	2	1	50.0%	Pharma / Industrial
2026-08-26	7	2	28.6%	Semiconductor / Software
2026-09-02	1	0	0.0%	Enterprise Tech
2026-09-09	2	2	100.0%	Consumer / Retail
2026-09-10	3	2	66.7%	Enterprise / Industrial

At an effective sample size of n = 6 cluster dates, the standard error on the hit rate is ±20.4pp. Dropping a single date (2026-08-26) swings headline accuracy from 51.9% to 60.0%.

1.3 Sample Size Power Calculation & Pre-Committed Stopping Rule

Detecting a true directional separation of δ = 2.0pp when idiosyncratic open-gap standard deviation σ ≈ 10.0pp (Effect Size d = 0.20) at α = 0.05 and Power 1 - β = 0.80 requires:

```
n_per_arm ≈ 2 * ((1.96 + 0.8416) / 0.20)^2 ≈ 392 observations per arm
Total N ≈ 784 independent observations (~157 weeks / 3 years at 5 prints/week)
```

Pre-Committed Stopping Rule: If the model is maintained in sandbox mode, enforce an explicit cutoff: If directional accuracy vs. the best constant rule yields an edge < +2.0pp at n = 100 total logged predictions (or

by **2027-02-15**), permanently decommission all directional scoring.

1.4 Governance Mechanics & Overfitting Audit

- **3-Confirmation Error Ledger Threshold:** Under pure noise where P(confirmation) = 0.5, the chance of 3 consecutive matching outcomes is 12.5%. Because failure categories are created dynamically by the author (unbounded multiple testing burden), the probability that at least one spurious category triggers 3 confirmations approaches 100%. The ledger acts as a noise-fitting engine.
- **Blindness & Selection Bias:** Clock guards and cryptographic hash locks prevent post-hoc changes (portfolio_aware: false). However, manual selection of earnings names introduces coverage selection bias.

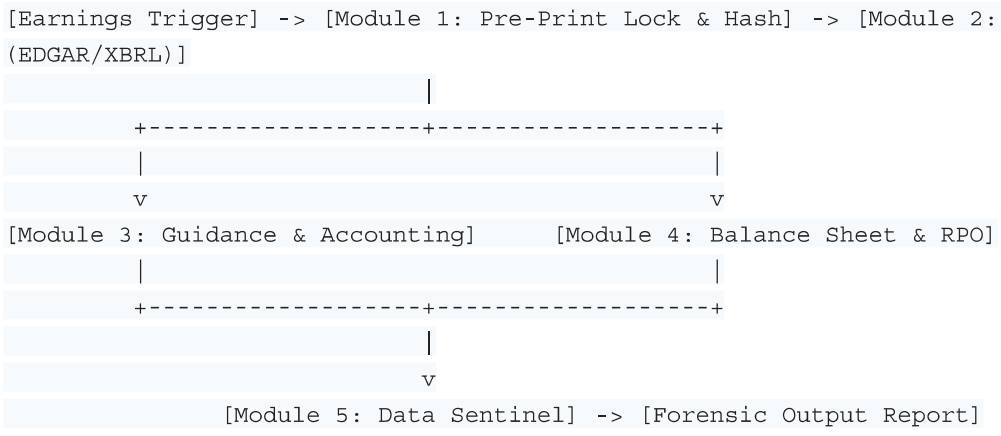
1.5 Execution Feasibility & Benchmark Contamination

- **Singapore Timezone Friction:** The US open (09:30 ET) occurs at 21:30 / 22:30 SGT. Observed gap decay is rapid (e.g., PLAB opened +27.1% and faded to +4% in 20 minutes; ORCL opened +7.51% and closed negative). Manual execution across SGT nights creates prohibitive slippage.
- **Mega-Cap Benchmark Contamination:** Calculating idiosyncratic return as R_idio = R_stock - β * R_sector fails when mega-caps move the ETF itself. ORCL moved IGV on its print night, distorting secondary names like ADBE.

Section 2: Technical Specification — PRE Earnings Quality & Forensic Engine

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2.1 Engine Architecture & Data Flow



2.2 Module Descriptions

- **Module 1: Pre-Print Lock & Clock Guard:** Enforces AMC (≤16:00 ET) and BMO (≤09:30 ET) cutoffs. Writes immutable SHA-256 pre-registration payload to audit_ledger.jsonl.
- **Module 2: Multi-Source Primary Ingestion:** Ingests 8-K (Item 2.02), 10-Q/K filings, IR press releases, and SEC XBRL statements within seconds of publication.
- **Module 3: Guidance & Accounting Reconciliation Engine (GARM):** Detects guidance basis shifts and low-quality earnings.
    - *Guidance Basis Shift:* ΔGuidance_clean = Guide_new - NonRecurringItems - Guide_old. (e.g. AEO $161M tariff refund inclusion converted a -4% underlying cut into a +36% reported raise).
    - *EPS Quality Ratio:* Q_eps = Operating Cash Flow / Net Income. Flag if Q_eps < 0.8 during an EPS beat.
- **Module 4: Balance Sheet & RPO Acceleration (XBRL Parser):** Tracks RevenueRemainingPerformanceObligation and Deferred Revenue. Measures RPO acceleration delta across consecutive quarters.
- **Module 5: Data Integrity & Calendar Sentinel:** Monitors vendor calendar drift (yfinance vs SEC filing dates) and flags zero-variance analyst feed corruption.

2.3 Forensic JSON Output Schema


```
{
  "audit_metadata": {
    "ticker": "AEO",
    "event_date": "2026-09-09",
    "integrity_hash":
"e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855",
    "post_cutoff": false
  },
  "forensic_flags": [
    {
      "category": "GUIDANCE_BASIS_SHIFT",
      "severity": "HIGH",
      "detail": "FY26 operating income raised $390-410M -> $540-550M. Inclusion of
$161M net tariff refund benefit masks baseline guidance cut of 4.0%.",
      "impact_score": -0.85
    },
    {
      "category": "CASH_FLOW_DIVERGENCE",
      "severity": "MEDIUM",
      "detail": "Net income increased 12% YoY while OCF decreased 4% YoY.",
      "impact_score": -0.40
    }
  ],
  "data_integrity": {
    "calendar_verified": true,
    "primary_source_matched": true,
    "feed_anomalies_detected": false
  }
```

}

Section 3: Probability Calibration & Brier Scoring Model

3.1 Benchmark Orthogonalization

To prevent mega-caps from distorting sector returns, calculate orthogonalized sector returns (R_sector\i):

```
R_sector\i = (W_sector * R_sector - w_i * R_i) / (W_sector - w_i)
R_idio* = R_stock - β_i * R_sector\i
```

3.2 Probabilistic Formulation & Brier Score Architecture

Model the breach probability p_breach = P(|R_idio*| > IV_implied). Evaluate calibration via Brier Score (BS):

```
BS = (1 / N) * ∑ (p_i - o_i)^2
```

Where o_i = 1 if |R_idio*| > IV_implied else 0. Uncalibrated coin-flip BS_null = 0.250; target performance BS < 0.200.

Brier Score Decomposition

```
BS = Reliability (Calibration Error) - Resolution (Discriminatory Ability) +
Uncertainty
```

3.3 Volatility Trading Decision Matrix

Parameter Condition	Trade Execution Strategy	Rationale
E[|R_idio*|] / IV_implied > 1.25 and **p_breach > 0.60**	**Buy Straddle / Long Volatility**	Options market is underpricing earnings jump risk.
E[|R_idio*|] / IV_implied < 0.75 and **p_breach < 0.35**	**Sell Short Iron Condor / Short Volatility**	Options market is overpricing earnings jump risk.

Parameter Condition	Trade Execution Strategy	Rationale
0.75 ≤ E[|R_idio*|] / IV_implied ≤ 1.25	**ABSTAIN (No Trade)**	No statistically significant volatility edge.

3.4 Python Implementation Code

```
import numpy as np

def compute_brier_score(predictions: np.ndarray, outcomes: np.ndarray) -> float:
    """Computes the Brier Score for probability calibration."""
    return float(np.mean((predictions - outcomes) ** 2))

def orthogonalized_idio_gap(r_stock: float, r_etf: float, w_stock: float, beta:
float) -> float:
    """Calculates idiosyncratic gap excluding stock's own contribution to ETF."""
    r_etf_ex_stock = (r_etf - w_stock * r_stock) / (1.0 - w_stock)
    return r_stock - beta * r_etf_ex_stock

# Example evaluation
implied_move = 0.08
actual_stock_gap = 0.12
actual_etf_gap = 0.02
stock_weight_in_etf = 0.15
beta = 1.2

idio_gap = orthogonalized_idio_gap(actual_stock_gap, actual_etf_gap,
stock_weight_in_etf, beta)
outcome = 1 if abs(idio_gap) > implied_move else 0
pred_prob = 0.68

brier = compute_brier_score(np.array([pred_prob]), np.array([outcome]))
print(f"Orthogonalized Idio Gap: {idio_gap:.4f}")
print(f"Breach Outcome: {outcome}, Brier Score Contribution: {brier:.4f}")
```

Summary Matrix

Component	Legacy PRE Model	Proposed Rescoped PRE System
Primary Goal	Binary Directional Prediction (UP/DOWN)	Forensic Quality-of-Earnings Audit + Volatility Edge
Core Metric	Binary Direction Hit Rate	Brier Score (BS) & Guidance Basis Flags
Benchmark	50% / Sector ETF Return	Implied Move (IV_implied) & Orthogonalized ETF
Sample Horizon	N ≈ 784 (Unachievable short-term)	N ≈ 50-100 (Achievable within months)
Value Driver	Directional trade signal	Accounting anomaly flags & magnitude edge